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Instruction

What the question is

My Notes:

TM Variants

Sketch a state diagram of a TM to recognize each of the languages described in: 3.8b (Assuming a leading blank)

UHH WE ALREADY DID THIS IN THE PREVIOUS ASSIGNMENT. AND IT HASN'T BEEN GRADED YET SO...??? I'm gonna copy and paste that because what????

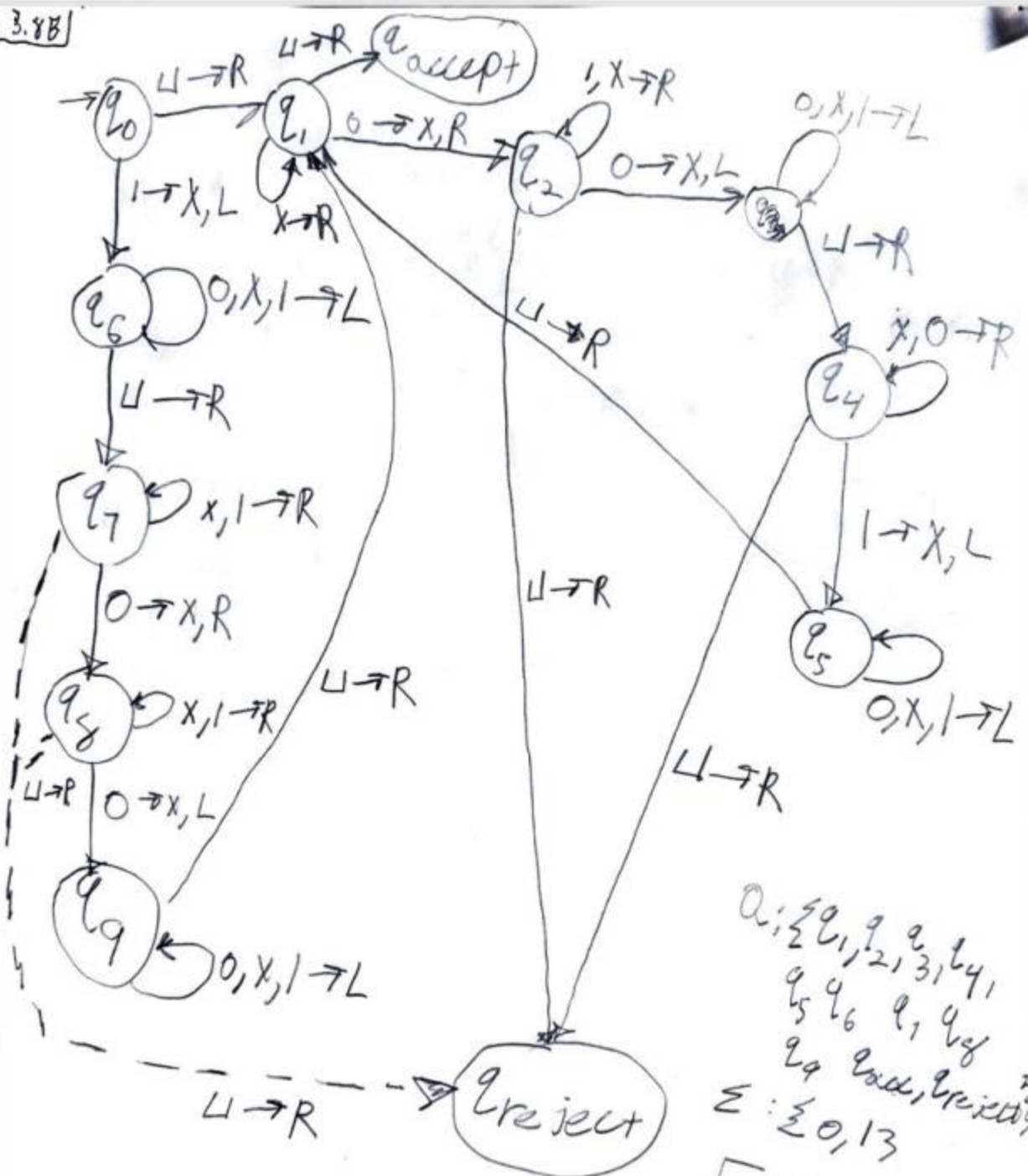
3.8 Give implementation-level descriptions of Turing machines that decide the following languages over the alphabet $\{0,1\}$.

a. $\{w \mid w \text{ contains twice as many 0s as 1s}\}$

b. $\{w \mid w \text{ contains twice as many 0s as 1s}\}$

c. $\{w \mid w \text{ contains twice as many 0s as 1s}\}$

3.8B



$Q: \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{\text{accept}}, q_{\text{reject}}\}$
 $\Sigma: \{0, 1\}$
 $\Gamma: \{0, 1, X, \sqcup\}$

$q_0: q_0$
 $q_{\text{accept}}: q_{\text{accept}}$
 $q_{\text{reject}}: q_{\text{reject}}$

$\delta: Q \times \Gamma \rightarrow Q \times \Gamma$
 $X \in L, R, \sqcup$

credit to Office hours since I didn't know where to start.

- 3.11** A *Turing machine with doubly infinite tape* is similar to an ordinary Turing machine, but its tape is infinite to the left as well as to the right. The tape is initially filled with blanks except for the portion that contains the input. Computation is defined as usual except that the head never encounters an end to the tape as it moves leftward. Show that this type of Turing machine recognizes the class of Turing-recognizable languages.

DID THIS PREVIOUSLY

3.11) To prove the double infinite tape T.M $D \equiv$ the ...
T.M M . To show equivalence between D & M . We need to
show 2 things

- Any Lang L that can be recognized by D can be recognized by M
- Any Lang L that can be recognized by M can be recognized by D as well.

1) Sim M by D

1) Mark left hand end of D

2) Prevent D from moving its head to the left of mark
 D sim M .

2) Sim $D \rightarrow M$

To sim the double tape Turing machine D by an
ordinary T.M M , sim it w/ a 2 tape T.M.

As the 2 tape T.M was already equiv is pow to
an ordinary T.M

Sim of L by 2 tape T.M D w/ 2 tape T.M M_1

1st tape of M_1 is written w/ input str &
the 2nd tape = blank

• Cut the tape of 2x 2 tape T.M D into parts, 2
at the starting cell of input str.

1 pt has input str & all blank space = right

2nd pt containing left of the input str
appears on the 2nd tape in reverse order.

M sim D .

Thus (1) & (2)

D is equiv to M . T.M w/ 2x tape is
equiv to an ordinary T.M

3.11 D.C From office hours

$$X \leq Y$$

$$Y \leq X$$

therefore

$$X = Y$$

① $\leftarrow \sqcup \text{a} \text{P P l e} \sqcup \sqcup \rightarrow$

$\text{a P P l e} \sqcup \dots$

② $\text{a P P l e} \sqcup \dots$

$\sqcup \text{a P P l e} \sqcup \dots$

You add a space for it. It proves that they have the same power. Therefore the double $\sqcup$ tape T.M recognizes the class of ~~TA~~ recog. Lang's

The T.M could ignore all the spaces left of the input & just do the same as a Normal T.M.

TM Closure: 3.15c, 3.15d, 3.15e

is Turing-recognizable.

3.15 Show that the collection of decidable languages is closed under the operation of

~~_____~~

~~_____~~

c. star.

d. complementation.

e. intersection.

AGAIN PREVIOUSLY DID ON OTHER ASSIGNMENT

3.156] Let L be a Turing decidable Lang. & M be the T.M that decides L .

There's a T.M M' s.t. that, it decides of L i.e., $L(M') = L^*$
Desc of M' is:

M' on input w :

1. Split w into n parts such that $w = w_1 w_2 \dots w_n$ in different ways.
2. Run M on w_i for $i = 1, 2, \dots, n$.

If M accepts each of these str $w_i \rightarrow$ accept

3. All cuts have been tried w/out success then reject.

When w is cut into differ substrings such that every str. is accepted by M , then w belongs to the str of L & thus M' accepts w after (finite # of steps), else w will $\rightarrow$ reject, b/c there are finitely many possible cuts of w , M' will halt after finitely many steps.

Therefore, $L(M') = L^*$. The decidable Lang. are closed under star.

DOUBLE CHECK:

3.15 L $w = w_1 \dots w_n$ on input w :



1. Split w into any # of pieces in all possible ways (≤ 1 of the pieces being ϵ).

2. Run D on each of the pieces

3. If all are accepted,

4. ^{accept} otherwise reject

3.15) (c) Let L be a decidable lang, decided by T.M. M .
 To decide L^* (the Kleene star of L , which is the set of strings formed by concat any finite # of str from L). We can construct a T.M. M' that operates as follows:

1. Given an input w , M' tries all possible ways to split w into k substr's $w = w_1 w_2 \dots w_k$ (for any $k \geq 0$). Where each w_i is a str that might belong to L .
2. For each split, M' simulates M on each substr w_i .
3. If M accepts each w_i in the split for any k , then M' accepts w .
4. If no such split exists, M' rejects w .

Since M halts on all inputs, M' will halt after trying all possible splits, so $L^* = \text{decidable}$

Inter section

$$L_1 \cap L_2 = \overline{\overline{L_1} \cup \overline{L_2}}$$

Both finite thus closed

From video

on

For 3.15e

3.15] For a Turing decidable Lang. L , T.M. decides
Lang. M then the complement is M' on input w .
The desc of M' is as follows

M' = on input w :

1. Accepts if M rejects
2. Else accept

Since M' does the opposite of what M does,
it decides the complement of L .

Therefore, decidable Lang. are closed under
complementation.

3.15e Let L_1 & L_2 be 2 Turing decidable Lang
 M_1 & M_2 be the T.M that decides L_1 & L_2 respectively
There is a T.M M' such that, it decides intersection
of L_1 & L_2 i.e. $L(M') = L_1 \cap L_2$
Desc of M'

M' on input w :

1. Run M_1 on w . If M_1 rejects then reject.
2. ELSE run M_2 on w . If M_2 rejects then reject
3. ELSE accept.

M' accepts w if both M_1 & M_2 accepts, if either
of the rejects then $M' \rightarrow$ rejects w .

Thus, $L(M') = L_1 \cap L_2$. The decidable Lang
are closed under intersect.

DOUBLE CHECK:

3.15 E) Let L_1 & L_2 be 2 decidable Lang's w/ T.M M_1 & M_2 that decide them. To decide the intersection $L_1 \cap L_2$, we can construct a T.M that simulates both M_1 & M_2 on the input w . The machine accepts w if both M_1 & M_2 accept w , & it rejects otherwise. Since M_1 & M_2 halt on all inputs, the machine for $L_1 \cap L_2$ will also halt on all inputs, making $L_1 \cap L_2$ decidable. Thus, the collection of decidable Lang's is closed under star, complement & intersection.

Credit:

For 3.15 C & E: Here are video that help my answer,

<https://www.youtube.com/watch?v=nZhAWPPDrF8>